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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 5

Team No.: 8

Project Title: High-Concurrency File Distribution Server

Team Members

Roll Number	Student Name	Signature
2520030278	Y.venkata sai	
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1. Abstract

The **High-Concurrency File Distribution Server** is a web-based system developed to make **file sharing faster and more reliable** when many users access files at the same time. The system allows authorized users to **upload, search, and download files** through a simple and user-friendly interface. The main focus of the project is to handle **multiple file requests simultaneously** using **Python and FastAPI** with asynchronous processing. It also provides a **resume download feature**, allowing users to continue a download from where it was interrupted instead of starting again.

An **admin monitoring dashboard** is included to keep track of **active users, downloads, server load, and transfer performance**. To improve security and reliability, the system uses **user authentication, access control, and SHA-256 file integrity verification**. **Postman** is used to test the APIs, while **Apache JMeter** is used to simulate multiple users and evaluate the server's **response time, throughput, and success rate**. The project also applies important **Operating Systems concepts**, including **processes, threads,**

concurrency, synchronization, memory management, file management, and Linux file I/O. Overall, the project aims to provide a **simple, secure, scalable, and efficient file distribution system** that can serve many users simultaneously.

2. Problem Statement

Traditional file-sharing servers can experience **increased response time** and **reduced download performance** when many users request files simultaneously. Large-file transfers may also fail when a connection is interrupted, forcing users to restart the download. **Managing multiple users efficiently** is another challenge for a basic file server. Our project addresses these limitations by developing a **High-Concurrency File Distribution Server** that can handle **multiple concurrent requests**, support **interrupted-download recovery**, and provide **basic server-performance monitoring**.

3. Objectives

- **server using processes and threads** to serve multiple users concurrently.
- **To implement concurrency and synchronization** for handling multiple file requests efficiently.
- **To efficiently manage memory and system resources** while transferring large files to multiple users.
- **To use Linux file systems and file I/O operations** for creating, reading, writing, and managing files.
- **To apply system calls and OS services** for efficient communication between the application and the operating system.

4. Proposed Methodology

- **Process & Thread Management :** The server uses **processes and threads** to handle multiple users and file requests at the same time without making users wait for each other.
- **Concurrency & Synchronization : Concurrent request handling** is used to manage multiple upload and download requests safely and avoid conflicts when users access shared files.
- **Memory Management :** The system uses **efficient memory and resource management** while transferring large files, helping the server work smoothly with multiple users.
- **File System & Linux File I/O :** The server uses **Linux file management and file I/O operations** to store, read, write, and transfer files efficiently.

- **System Calls & OS Services :** The application uses **system calls and operating-system services** to communicate with the underlying Linux system for file and resource operations.
- **Performance Testing & Monitoring: Postman** is used to test the APIs, while **Apache JMeter** is used to simulate multiple users and measure server performance such as **response time, throughput, and successful requests**.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API	Application in the Project
Processes and Threads (CO-2)	Concurrent server workers/requests and independent task handling.
Process Synchronization & Concurrency (CO-6)	Safe handling of simultaneous file requests and shared resources.
File Systems & File Abstractions (CO-5)	Create, read, write, seek, store, and manage distributed files.
Linux File I/O — open(), read(), write(), close()	Low-level interaction with files during upload/download operations.
Memory Management (CO-4)	Efficient buffering and controlled memory use for large file transfers.
Virtual Memory / Page Fault awareness (CO-4)	Understanding memory pressure while serving large files and many requests.
System Calls & Kernel Services (CO-1)	User-space server interacts with OS services through system-call abstractions.
Semaphores / Locks (CO-6)	Protect shared metadata/resources from race conditions when required.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030278	Y.venkata sai	
2520030318	B.Ram Charan	
2520030407	D.Tejas	

7. Tools / Platforms / Software Used

Tool / Platform	Purpose
VS Code	Development and source-code editing
Python + FastAPI	Backend server and concurrent request handling

HTML, CSS, JavaScript	Web interface
MySQL	User, file metadata and download-history storage
Postman	API testing
Apache JMeter	Concurrent-load and performance testing
Linux / Ubuntu	Operating-system environment and Linux file I/O
Git & GitHub	Version control and project repository

8. Expected Outcome

- **System Calls & OS Services:** The system is expected to use **OS services and system calls** for efficient file and resource operations.
- **Processes & Threads:** The server is expected to handle **multiple users simultaneously** using processes and threads, improving concurrent file distribution.
- **Memory Management:** The system is expected to use **memory and system resources efficiently** while transferring large files and serving multiple users.
- **File Systems:** The system is expected to provide **efficient file management and Linux file I/O**, including reading, writing, storing, and transferring files.
- **Concurrency & Synchronization:** The system is expected to safely manage **concurrent file requests and synchronization**, reducing conflicts and improving server performance.
- **Overall Outcome:** Performance testing with **Apache JMeter** is expected to demonstrate the server's **response time, throughput, and request success rate** under increasing numbers of simultaneous users.

9. GitHub Repository

GitHub Repository Link: https://github.com/yerramasuvenkatasai-cell/ossp-project

Repository Created:

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name:

Signature:

Remarks:

Faculty Use

Project Approved:

☐ Yes ☐ No

Date:

Faculty Signature: